

# An Agricultural Revolution Is Unfolding in India

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In recent years, a quiet agricultural revolution has been unfolding in Andhra Pradesh, a state in southeastern India. Once known as one of India's rice bowls, the region has faced soil degradation, water stress and farmer debt after years of heavy reliance on chemical fertilizers, pesticides and deep tillage.

In response, the Government of Andhra Pradesh launched Andhra Pradesh Community Managed Natural Farming (APCNF) in 2016. Its goal is ambitious: by 2031, to help 6 million farmers and 6 million hectares of farmland transition to natural farming, making it one of the largest agroecology programs in the world.

By October 2023, APCNF had reached 3,730 villages across 26 districts, involving 851,000 farming households and covering 278,000 hectares. About 90% of participating farmers are smallholders.

Why this case matters: As extreme weather becomes more frequent, old methods alone are no longer enough. Learning from agricultural experiences in other regions can expand our thinking, open new paths and strengthen the climate resilience of farming.

## The Core Idea of Natural Farming



Natural farming uses local resources and soil life instead of chemical inputs.

APCNF is built on the idea of working with nature. Instead of chemical fertilizers and pesticides, it uses local resources - cow dung, urine, crop residues, plant extracts and other biological materials - to restore soil health. Its principles are simple: imitate natural ecosystems and allow the land to recover its own vitality.

## Nine Principles: Farming the Way Nature Works

- Keep living roots in the soil year-round: the land is never left bare, and plant roots continuously provide carbon and energy to soil life.
- Diversify cropping: at least 15 to 20 crops, and sometimes trees, grow together in one field, forming a stable mini-forest.
- Reduce tillage or practice no-till: minimize disturbance to protect soil structure and organic matter.
- Integrate crops and livestock: animals and crops support one another, returning manure, compost and other organic matter to the soil.
- Use biological preparations: cow dung, urine and fermented plant extracts help activate soil microbial communities.
- Cover the soil with residues: when crops are not growing, straw or other residues protect the soil from drying, cracking and carbon loss.
- Prioritize local seed: farmer-saved and locally adapted varieties are better suited to local climate conditions.
- Manage pests ecologically: integrated measures and plant-based extracts replace synthetic pesticides.
- Eliminate chemical fertilizers and pesticides: chemical inputs are phased out so that the land can rebuild its own cycling capacity.

## Results: Better Income, Healthier Soil and Stronger Communities



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Over the past several years, research teams have conducted crop-cutting experiments on more than 3,000 fields, comparing natural farming with chemical agriculture. The results are encouraging.

- Higher farm income: 77% of farmers saw production costs fall, while 67% reported higher income. Average net profit per hectare was about 27% higher than in chemical farming. A GIST study also found that natural farming households earned 49% more net income, while chemical inputs fell by 56% to 73%.
- Healthier soil: cover crops and no-till practices reduce carbon loss, increase microbial diversity and gradually raise soil organic carbon.
- Greater resilience to extreme weather: during droughts and heavy rain, natural farming plots retain more water and suffer less crop loss. Diverse planting systems make farmland more resilient.
- Stronger carbon benefits: agroforestry and cover crops increase carbon sequestration. Reduced tillage also lowers carbon emissions and improves soil fertility.
- Improved health: farmers have less pesticide exposure, workdays lost to illness fall by 30%, and household diets become more diverse.
- Stronger community and environment: village cohesion improves, farmers return to their communities, women's self-help groups become more active, and smallholders benefit significantly.
- Lower market dependence: natural farming households rely more on local sales and self-provisioning, making them less vulnerable to external price fluctuations.

## A Farmer's Story and the Future of APCNF





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One video in the original article tells the story of Imam Kasim, a young man who returned to his village in India after working in the city. With guidance from APCNF, he adopted natural farming, intercropping and fully natural biological preparations. He reduced production costs, improved crop yields and secured a stable income. Kasim has since enrolled in the Farmer Scientist course at the Indo-German Global Academy for Agroecology Research and Learning (IGGAARL) in Pulivendula, Andhra Pradesh.

The Food and Agriculture Organization of the United Nations (FAO) and international researchers predict that if Andhra Pradesh fully transitions to natural farming by 2050, the state could provide more food per person per day than chemical agriculture, with diets that are more diverse and nutritious. Without chemical fertilizers and pesticides, soils would become increasingly fertile and rural areas would gain more diverse employment opportunities.

## Conclusion



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Natural farming in Andhra Pradesh is not a return to the primitive past. It is a meeting point between science and traditional knowledge - a way of farming that can nourish farmers, soil and the future at the same time.

For China and for the wider world as they search for green transitions, this case offers a valuable lesson: agriculture is not only about producing food. It is also about learning to cooperate with nature.

## References and Resources

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4. Impact Assessment Report 2022-23 | IDS Andhra Pradesh - Andhra Pradesh Community Managed Natural Farming.

## About the Soil Vitality Lab

The Soil Vitality Research and Enhancement Laboratory at Sichuan University was established in 2021. Its core members include Dr. Hongyan Lu, Dr. Jiong Yan, Dr. Xue Chen and Ms. Wei Tang. The lab promotes dialogue between field practice and science, supporting farmers as they explore localized pathways to rebuild soil vitality and strengthen agricultural climate resilience.